Time and Ordering II

Material based on Lectures by Indy Gupta(IG) - All slides © IG

Reading

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- 14.4 in Coulouris et al.
- 6.2 in Tanenbaum

Next

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- Can we avoid time-synchronization?
- Logical Clocks: Lamport Time Stamps

Ordering Events in a Distributed System

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- To order events across processes, trying to sync clocks is one approach
- What if we instead assigned timestamps to events that were not *absolute* time?
- As long as these timestamps obey causality, that would work

If an event A causally happens before another event B, then timestamp(A) < timestamp(B)

- Humans use causality all the time
  - E.g., I enter a house only after I unlock it
  - E.g., You receive a letter only after I send it

Logical (or Lamport) Ordering

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- Proposed by Leslie Lamport in the 1970s
- Used in almost all distributed systems since then
- Almost all cloud computing systems use some form of logical ordering of events

Logical (or Lamport) Ordering(2)

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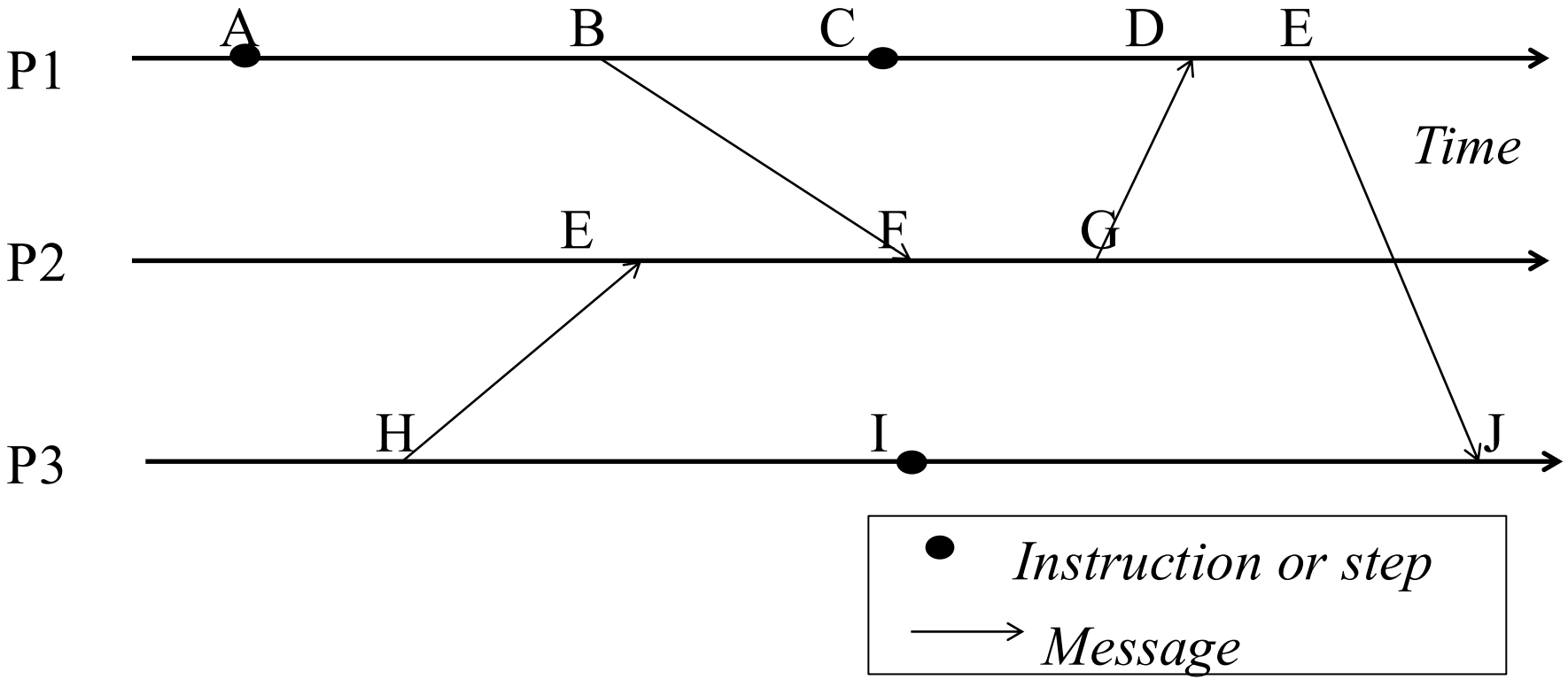
- Define a logical relation *Happens-Before* among pairs of events

- Happens-Before denoted as $\rightarrow$

- Three rules
  1. On the same process: $a \rightarrow b$, if $time(a) < time(b)$ (using the local clock)
  2. If p1 sends *m* to p2: $send(m) \rightarrow receive(m)$
  3. (Transitivity) If $a \rightarrow b$ *and* $b \rightarrow c$ then $a \rightarrow c$

- Creates a *partial order* among events
  - Not all events related to each other via $\rightarrow$

Example



Happens-Before

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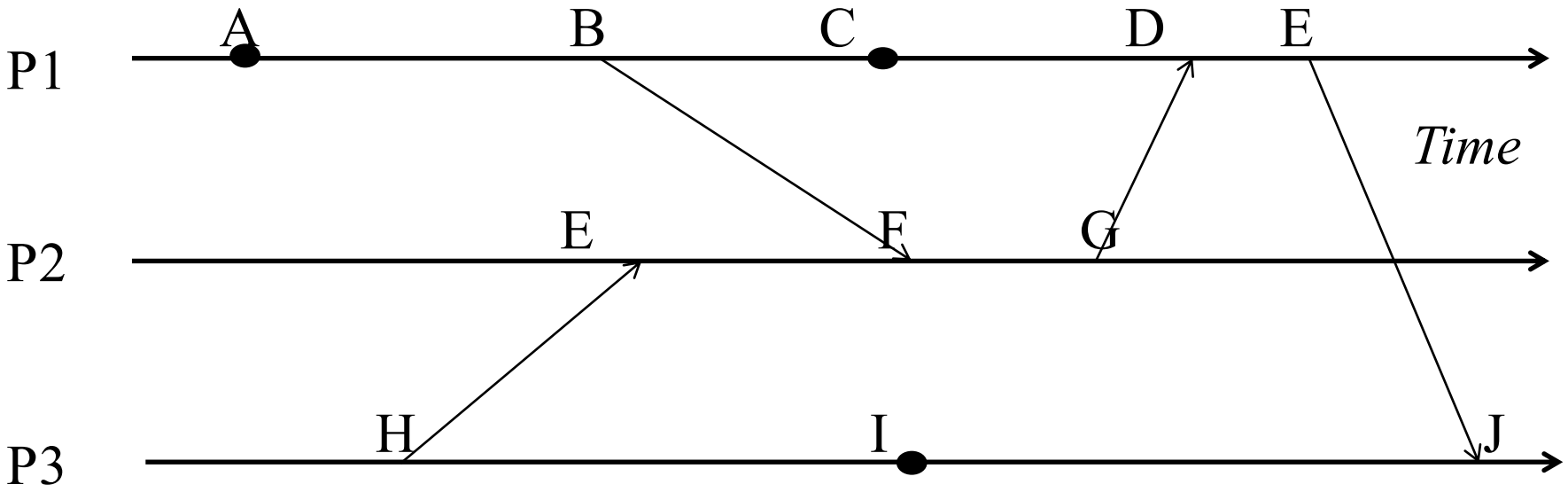

- A → B
- B → F
- A → F

● *Instruction or step*

→ *Message*

Happens-Before

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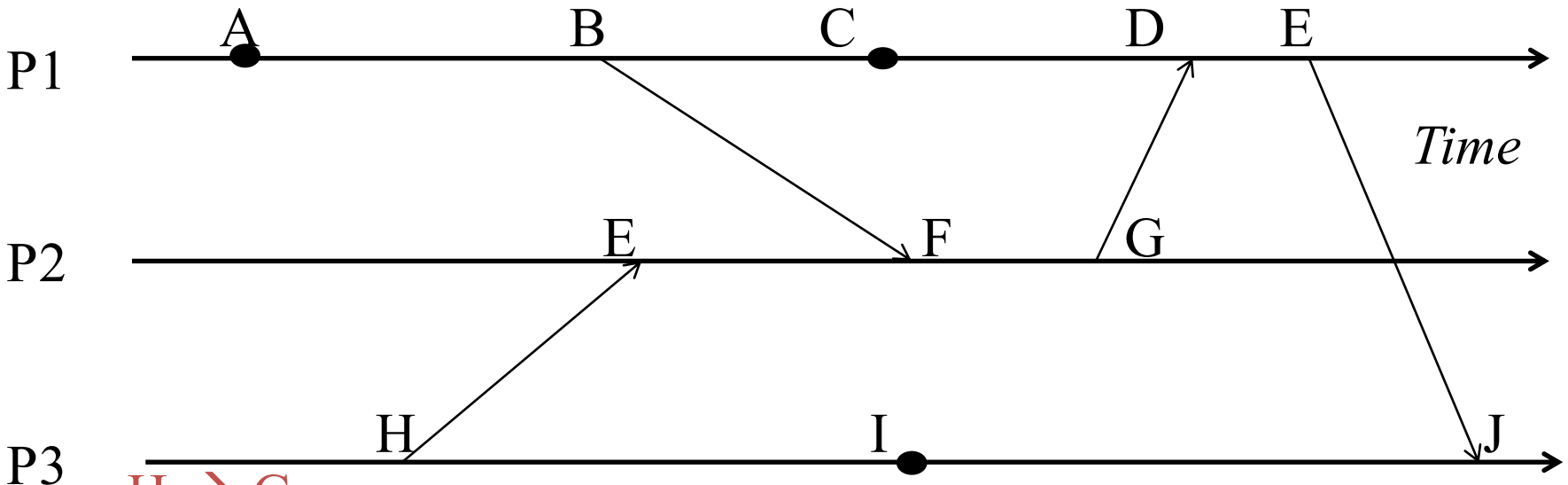

- H → G
- F → J
- H → J
- C → J

● *Instruction or step*

→ *Message*

In practice: Lamport timestamps

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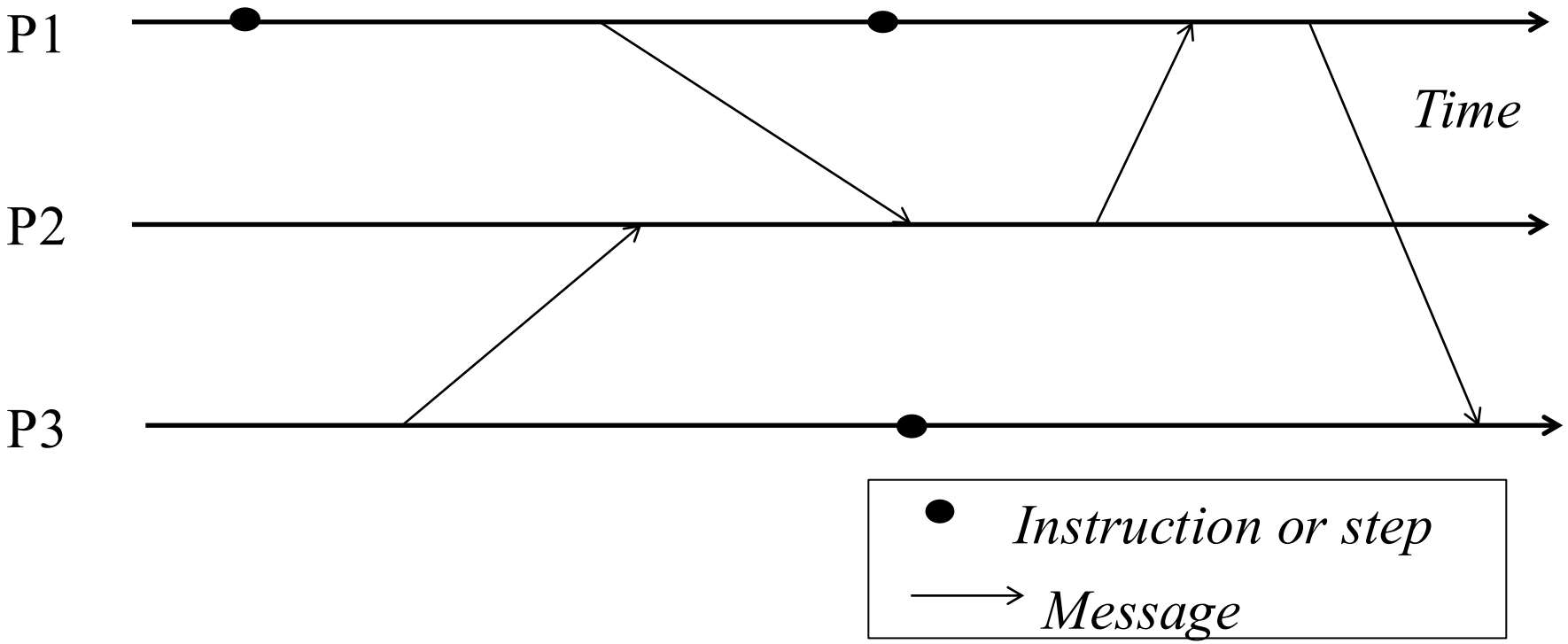
- Goal: Assign logical (Lamport) timestamp to each event

- Timestamps obey causality

- Rules
  - Each process uses a local counter (clock) which is an integer
    - **initial value** of counter is **zero**
  - A process increments its counter when a send or an instruction happens at it. The counter is assigned to the event as its timestamp.
  - A send (message) event carries its timestamp
  - For a receive (message) event the counter is updated by

    max(local clock, message timestamp) + 1

Example



Lamport Timestamps

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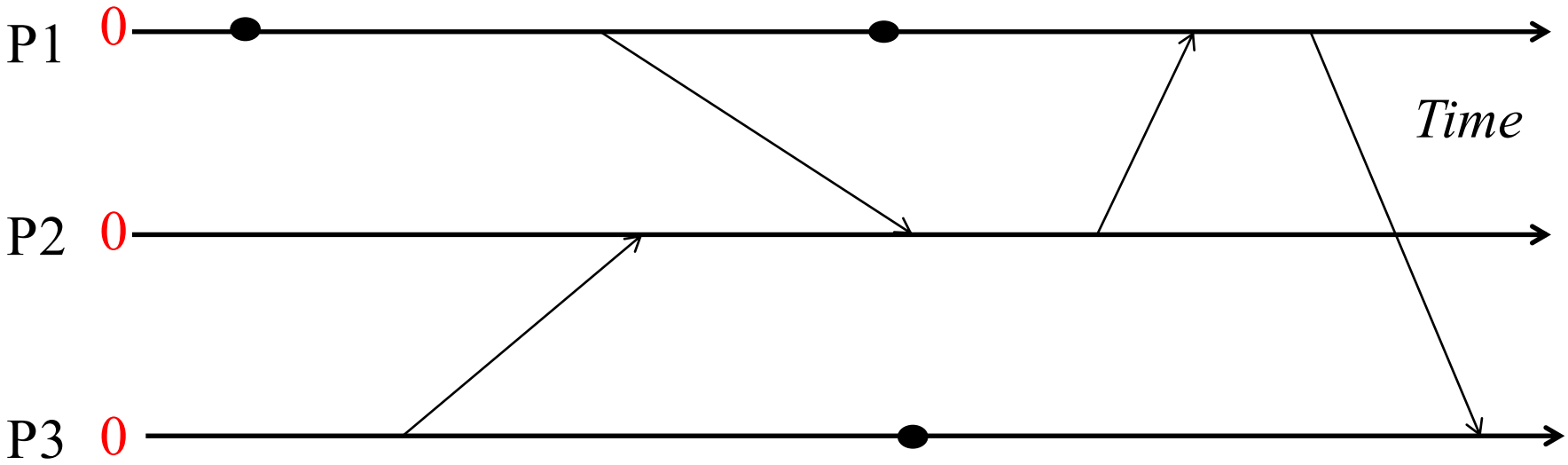
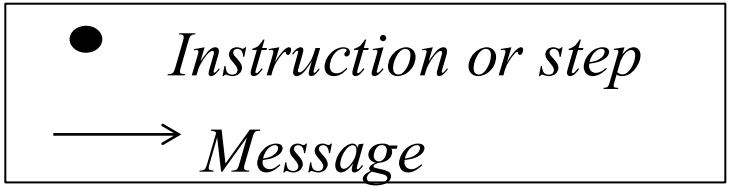


Initial counters (clocks)



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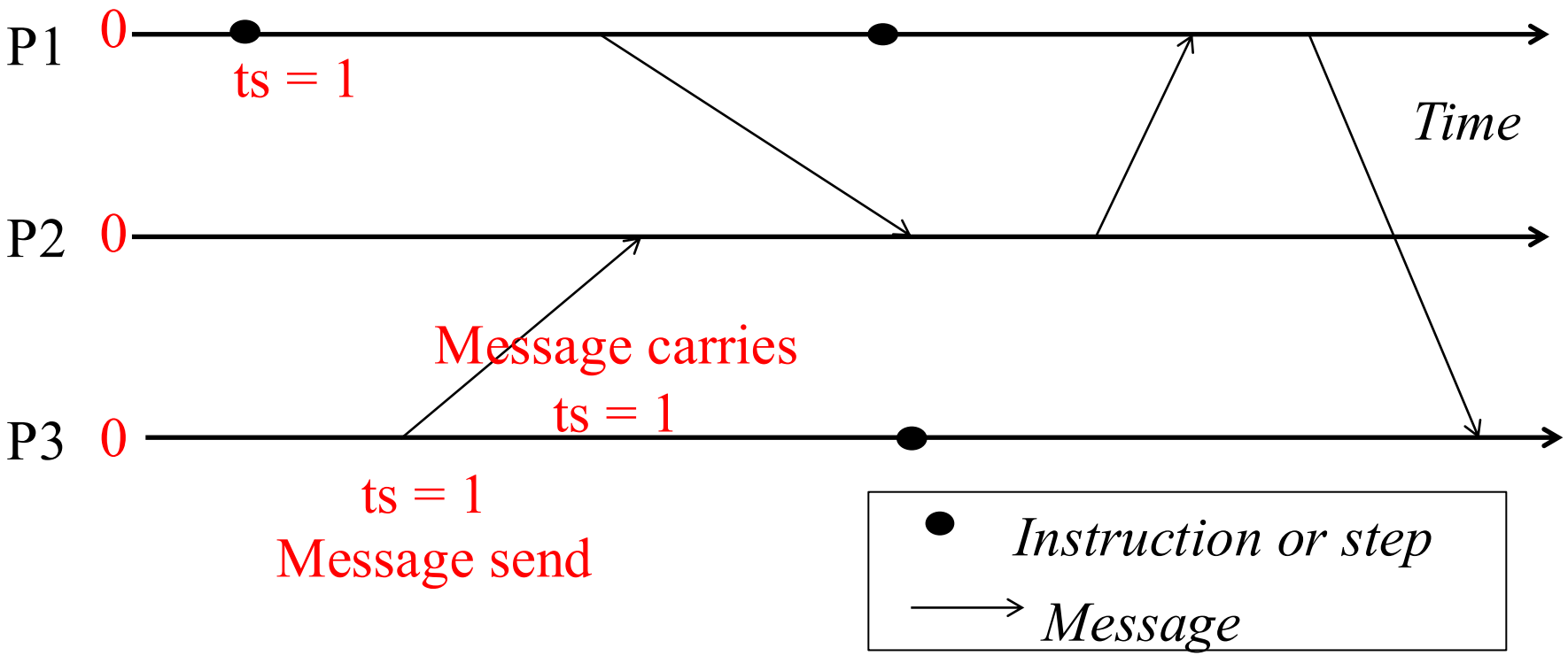

Lamport Timestamps

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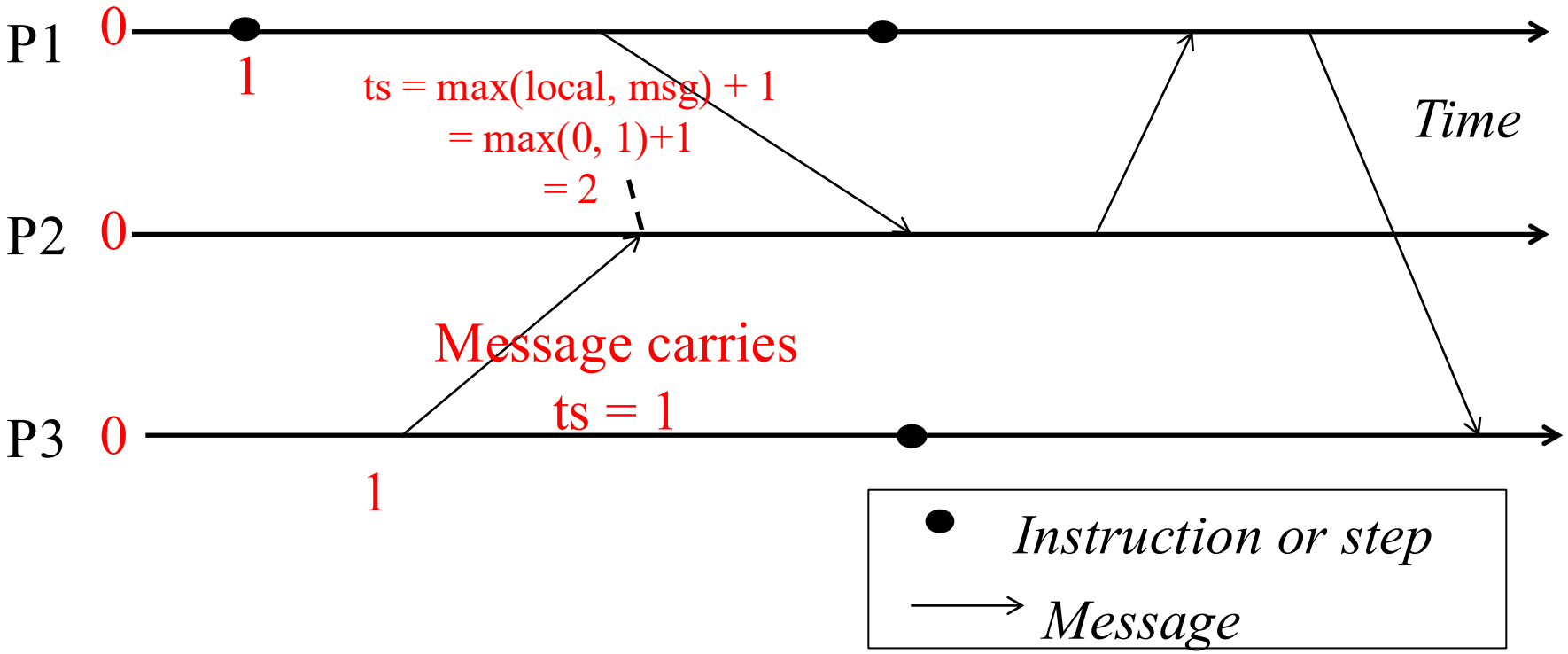

Lamport Timestamps

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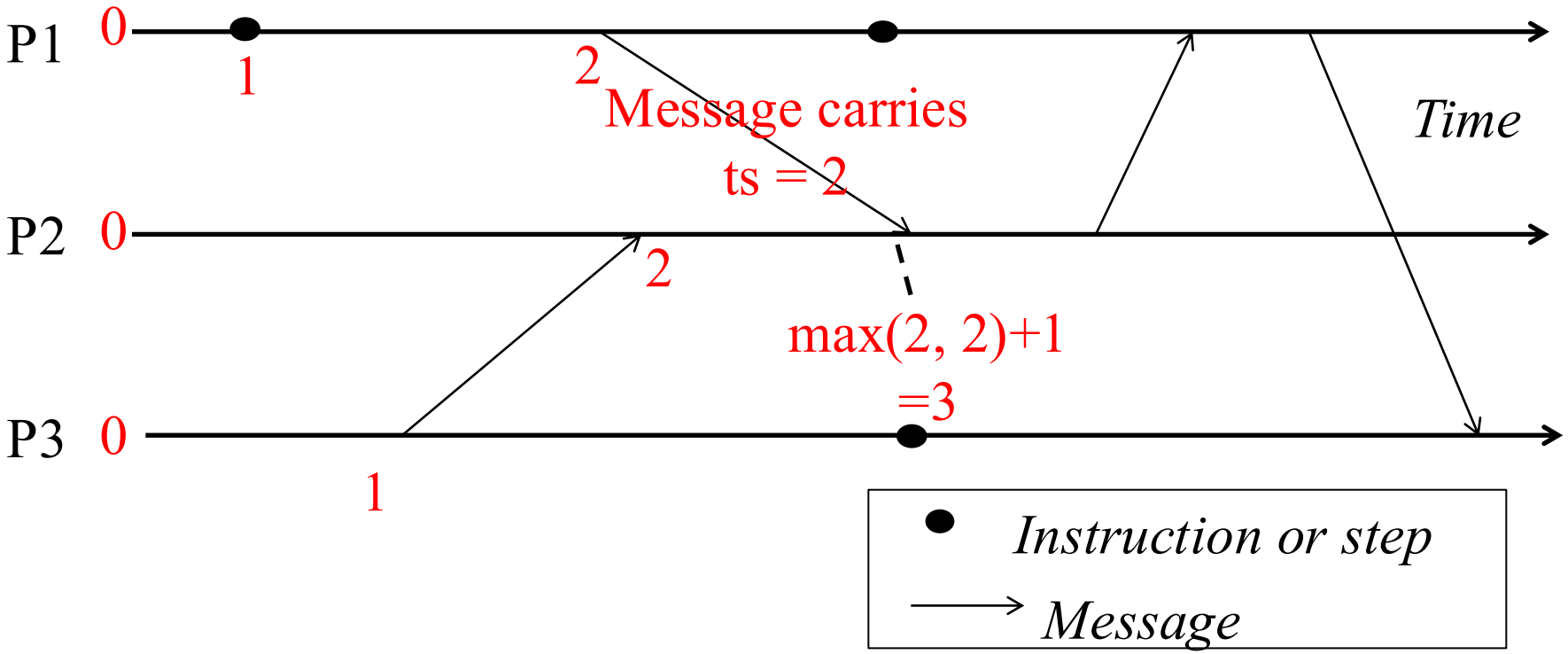

Lamport Timestamps

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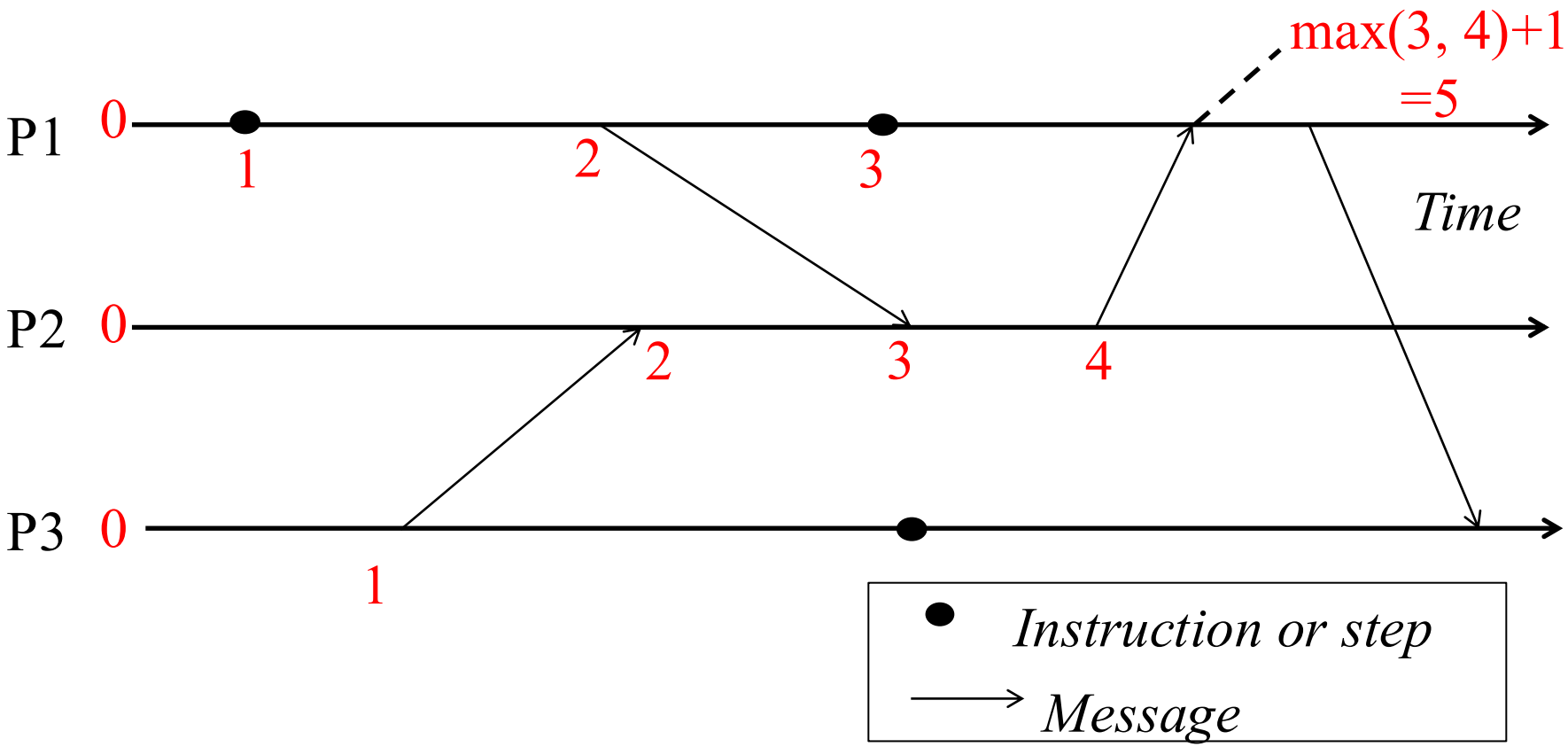

Lamport Timestamps

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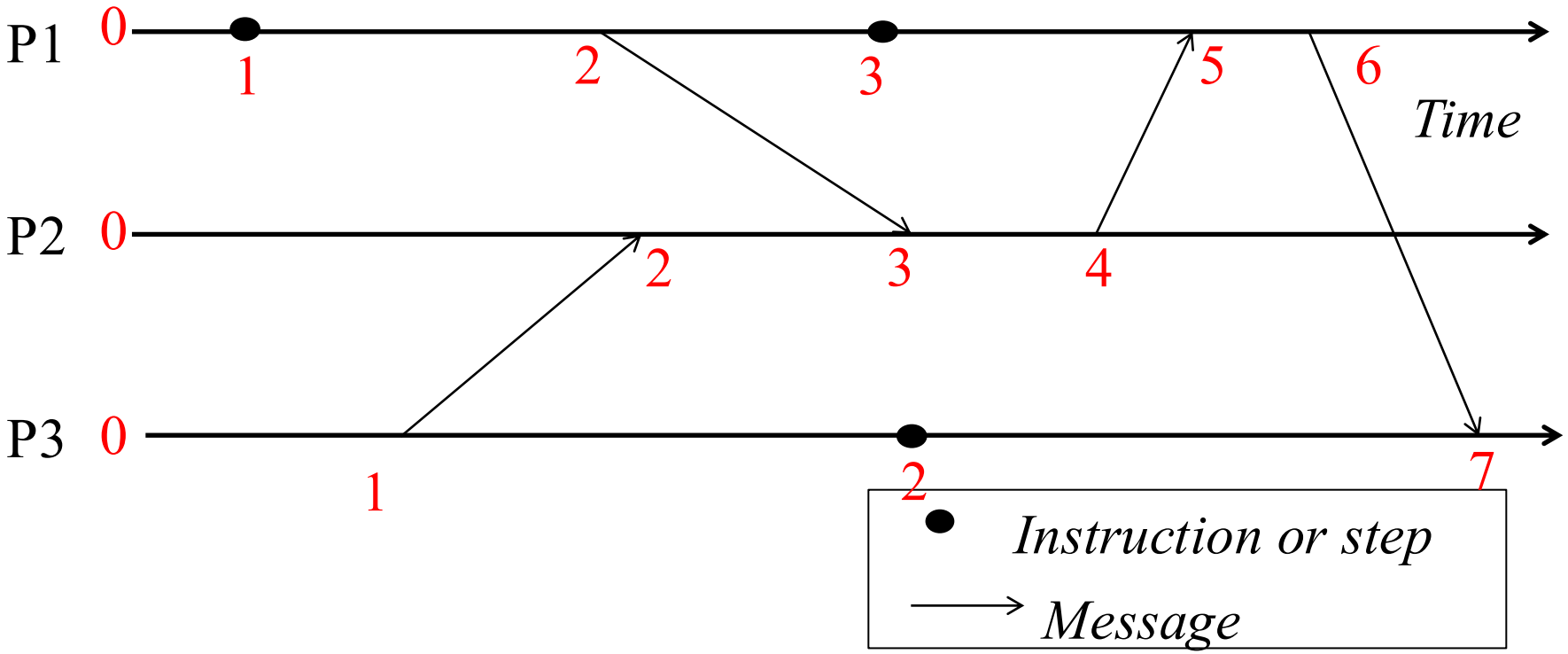

Obeying Causality

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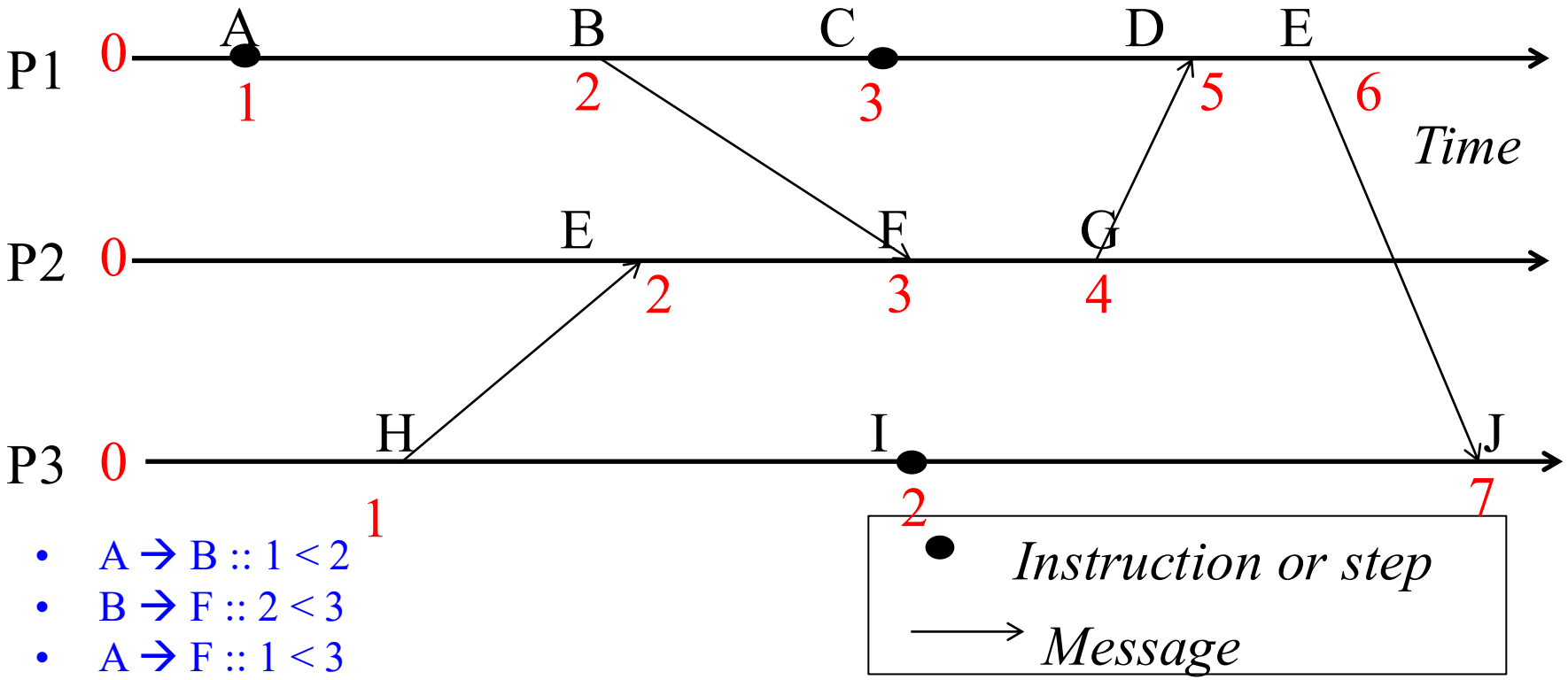

- A → B :: 1 < 2
- B → F :: 2 < 3
- A → F :: 1 < 3

Obeying Causality (2)

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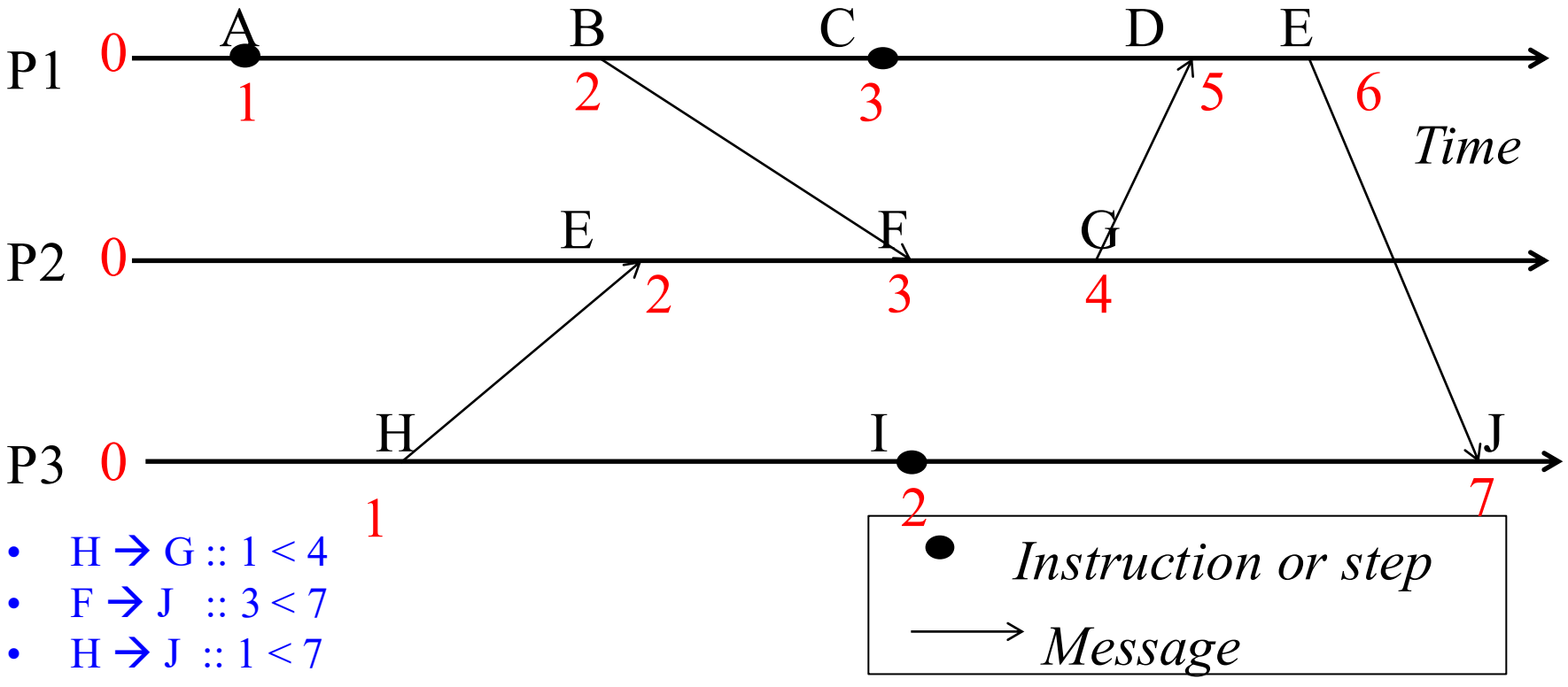

- H → G :: 1 < 4
- F → J :: 3 < 7
- H → J :: 1 < 7
- C → J :: 3 < 7

Not always *implying* Causality

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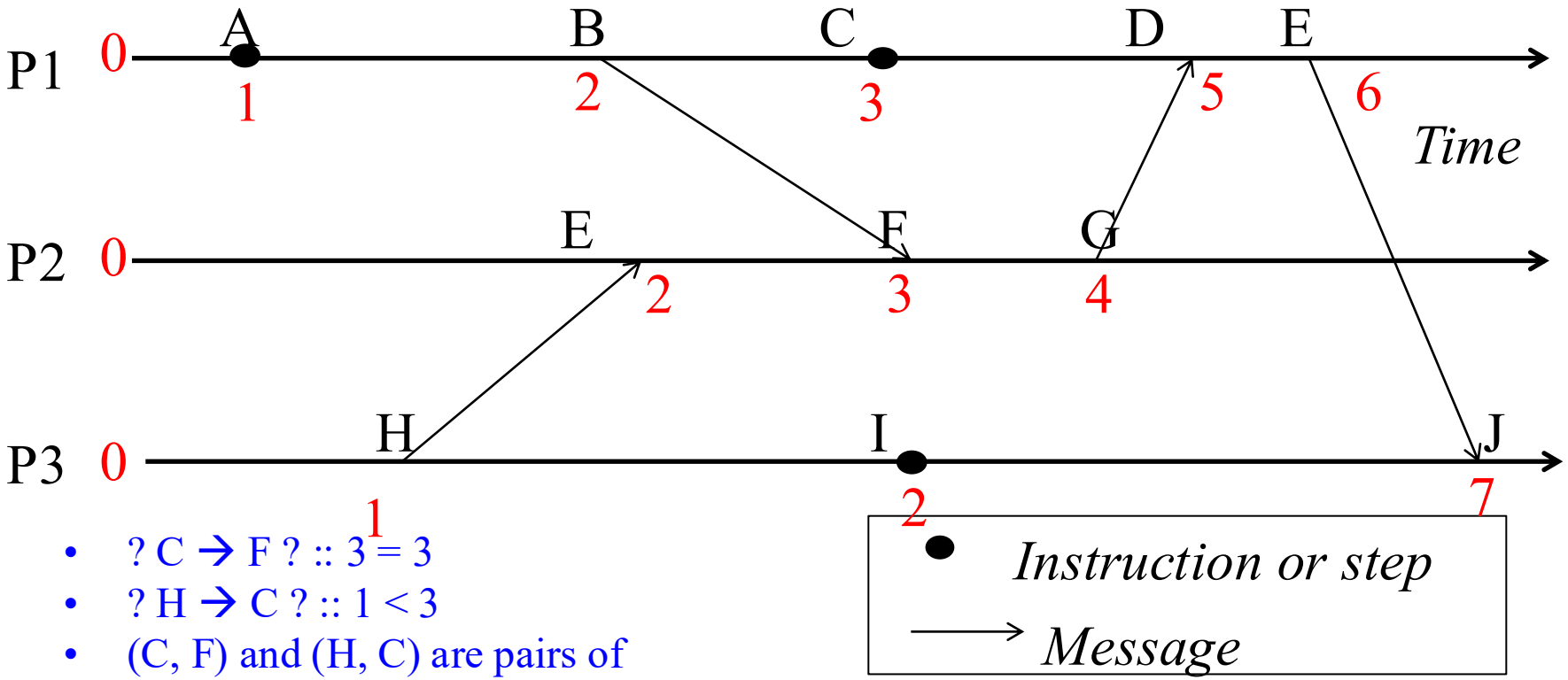

- ? C → F ? :: 3 = 3
- ? H → C ? :: 1 < 3
- (C, F) and (H, C) are pairs of *concurrent* events

Concurrent Events

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- A pair of concurrent events doesn't have a causal path from one event to another (either way, in the pair)

- Lamport timestamps not guaranteed to be ordered or unequal for concurrent events

- Ok, since concurrent events are not causality related!

- Remember

  E1 → E2 ⇒ timestamp(E1) < timestamp (E2), BUT

  timestamp(E1) < timestamp (E2) ⇒

  {E1 → E2} OR {E1 and E2 concurrent}

Next

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Can we have causal or logical timestamps from which we can tell if two events are concurrent or causally related?

Next

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- Algorithms for Clock Synchronization
- Logical Clocks: Vector Clocks

Vector Timestamps

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- Used in key-value stores like Riak
- Each process uses a vector of integer clocks
- Suppose there are N processes in the group 1…N
- Each vector has N elements
- Process *i maintains vector* $\mathbf{V_i[1...N]}$
- *j*th element of vector clock at process *i*, $V_i[j]$, is *i*'s knowledge of latest events at process *j*

Assigning Vector Timestamps

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Incrementing vector clocks

1. On an instruction or send event at process *i*, it increments only its *i*th element of its vector clock

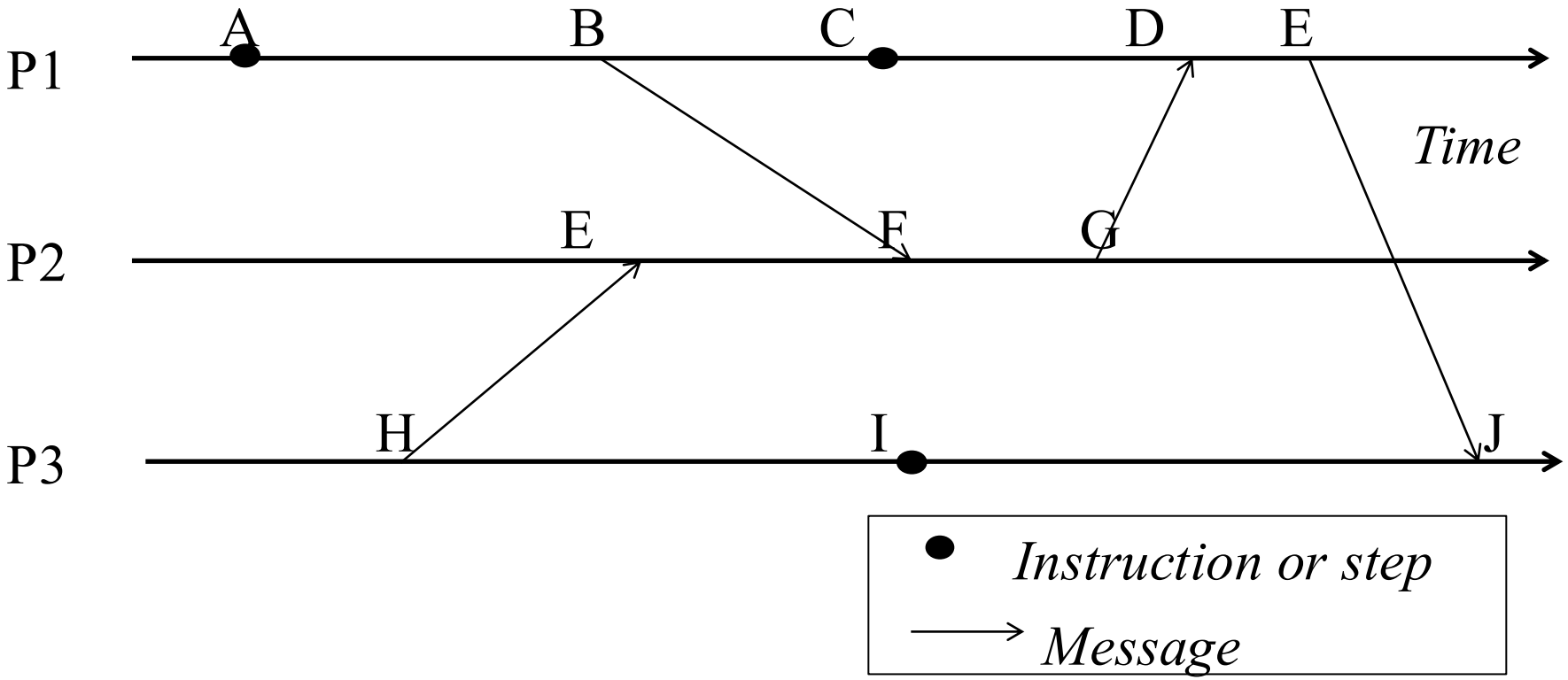
2. Each message carries the send-event's vector timestamp $V_{message}[1 \ldots N]$

3. On receiving a message at process *i*:

   $\mathbf{V_i[i] = V_i[i] + 1}$

   $\mathbf{V_i[j] = \max(V_{message}[j], V_i[j])}$ **for** $\mathbf{j \neq i}$

Example



Vector Timestamps

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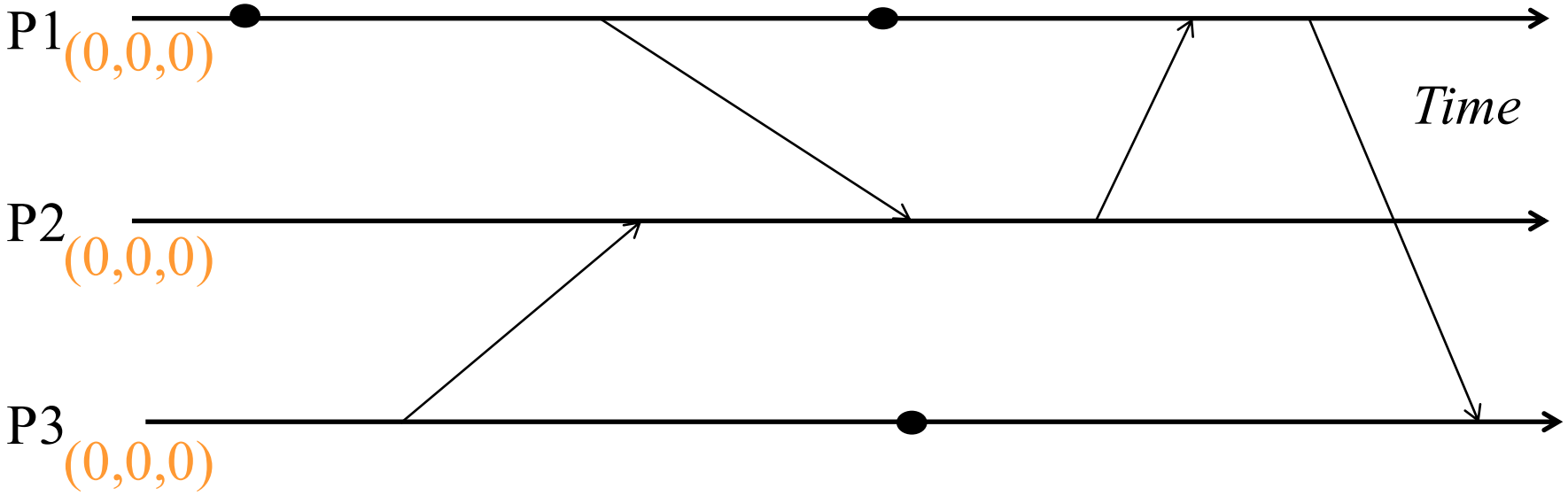

Initial counters (clocks)

Vector Timestamps

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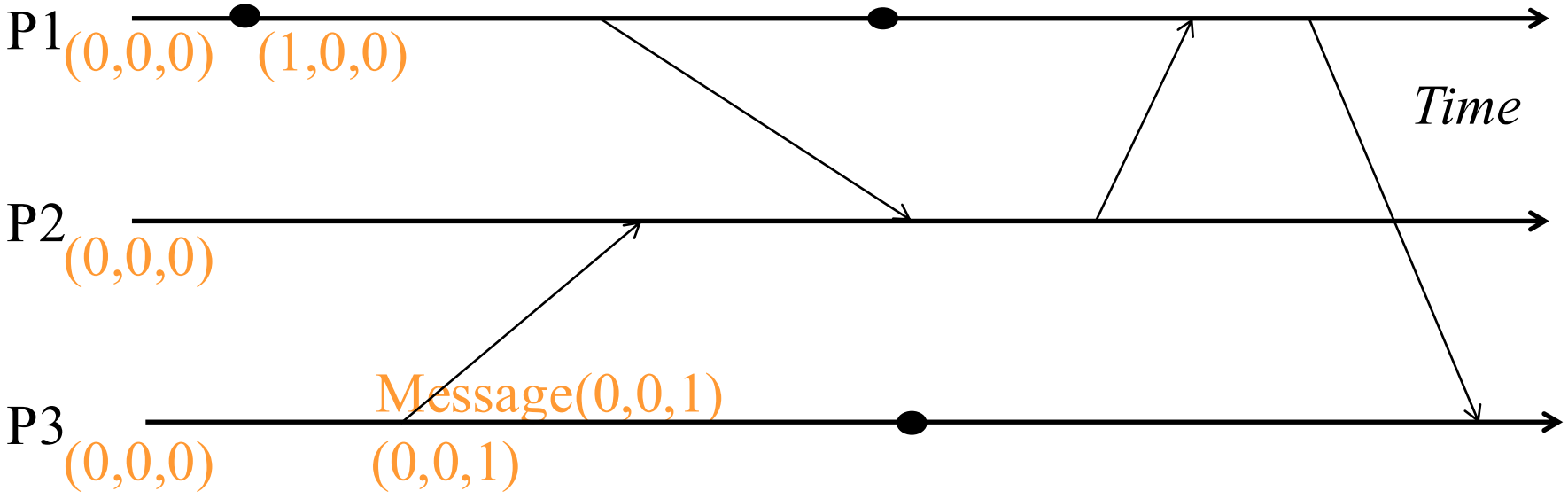

Vector Timestamps

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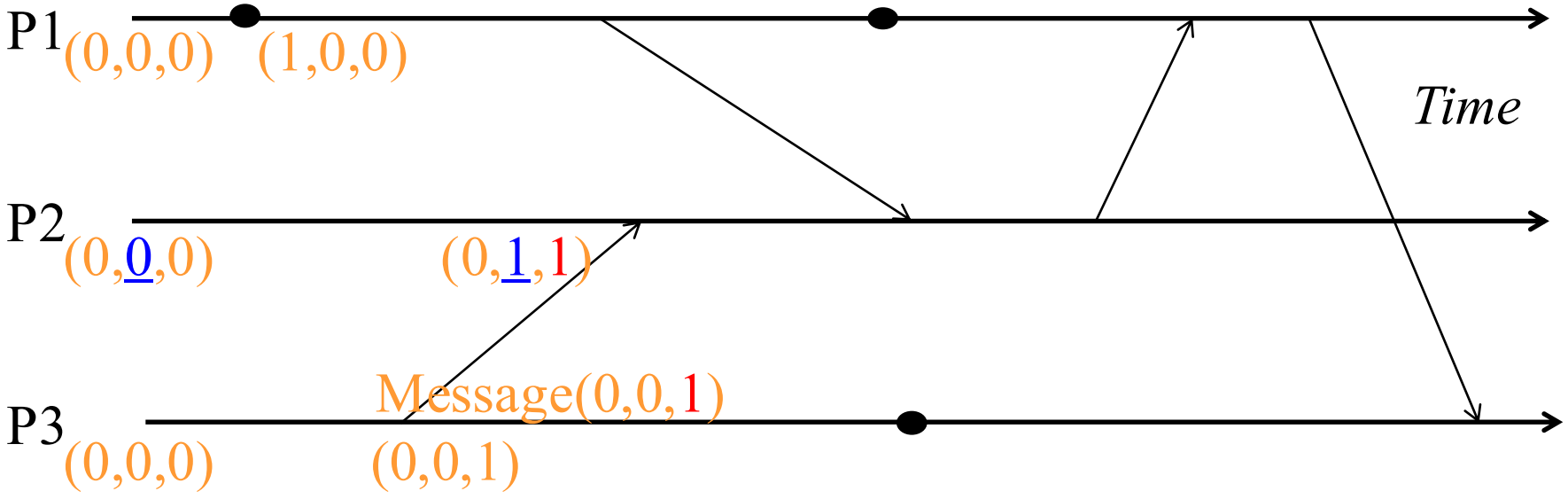

Vector Timestamps

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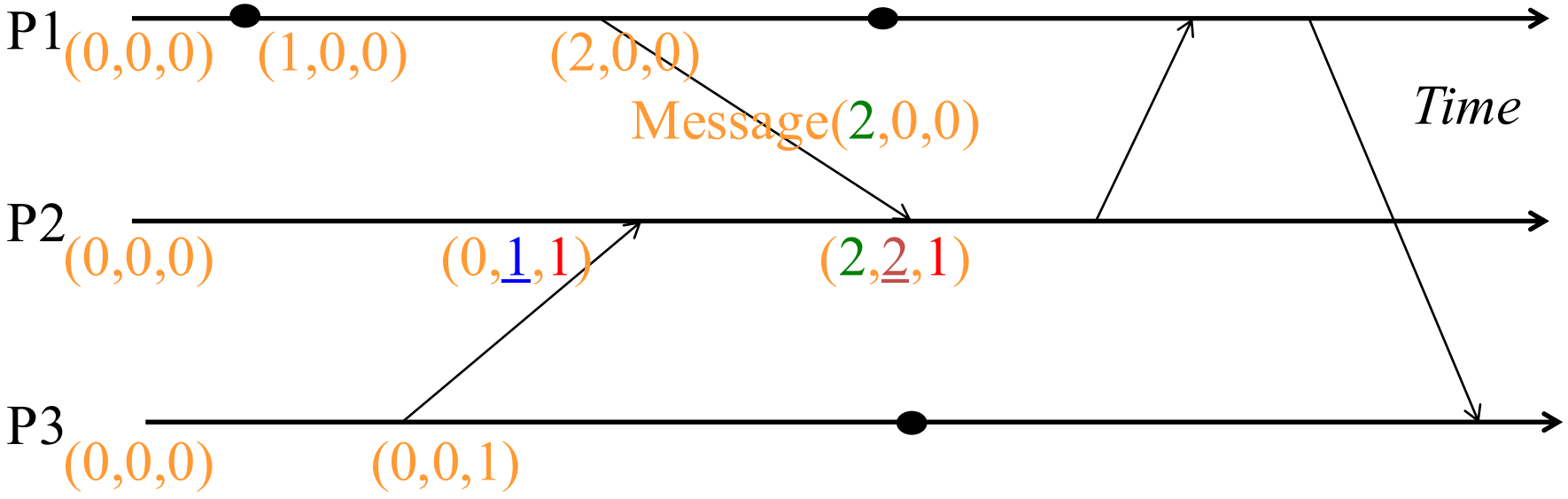

Vector Timestamps

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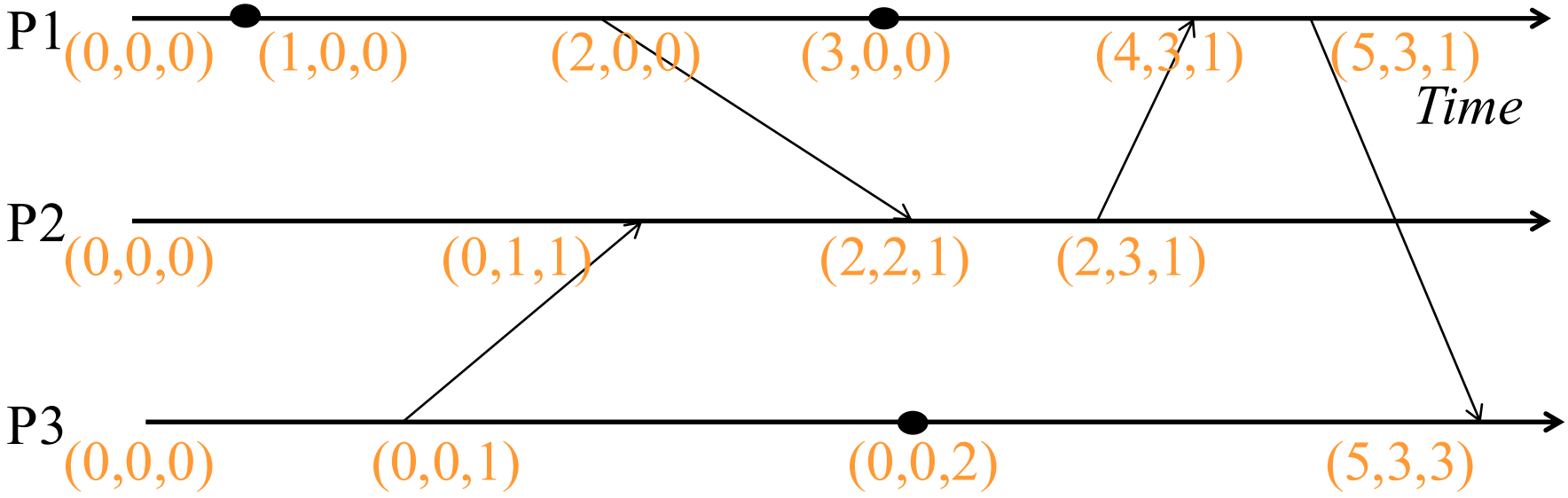

Causally-Related …

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- $VT_1 = VT_2$, *iff* (if and only if) $VT_1[i] = VT_2[i]$, for all $i = 1, \ldots, N$
  - EX: (0,0,0) = (0,0,0)

- $VT_1 \leq VT_2$, *iff* $VT_1[i] \leq VT_2[i]$, for all $i = 1, \ldots, N$
  - EX: (0,0,0) ≤ (0,0,0)
  - EX: (3,2,1) ≤ (3,2,2)

- Two events are causally related *iff* $VT_1 < VT_2$,
  i.e., *iff* $VT_1 \leq VT_2$ & there exists j such that $1 \leq j \leq N$ & $VT_1[j] < VT_2[j]$
  - EX: (3,2,1) < (3,2,2)

… or Not Causally-Related

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- Two events VT_1 and VT_2 are concurrent *iff*

$$\text{NOT } (VT_1 \leq VT_2) \text{ AND NOT } (VT_2 \leq VT_1)$$

We'll denote this as $VT_2 \mid\mid\mid VT_1$

- EX: (4,2,1) ||| (3,2,2)

Obeying Causality

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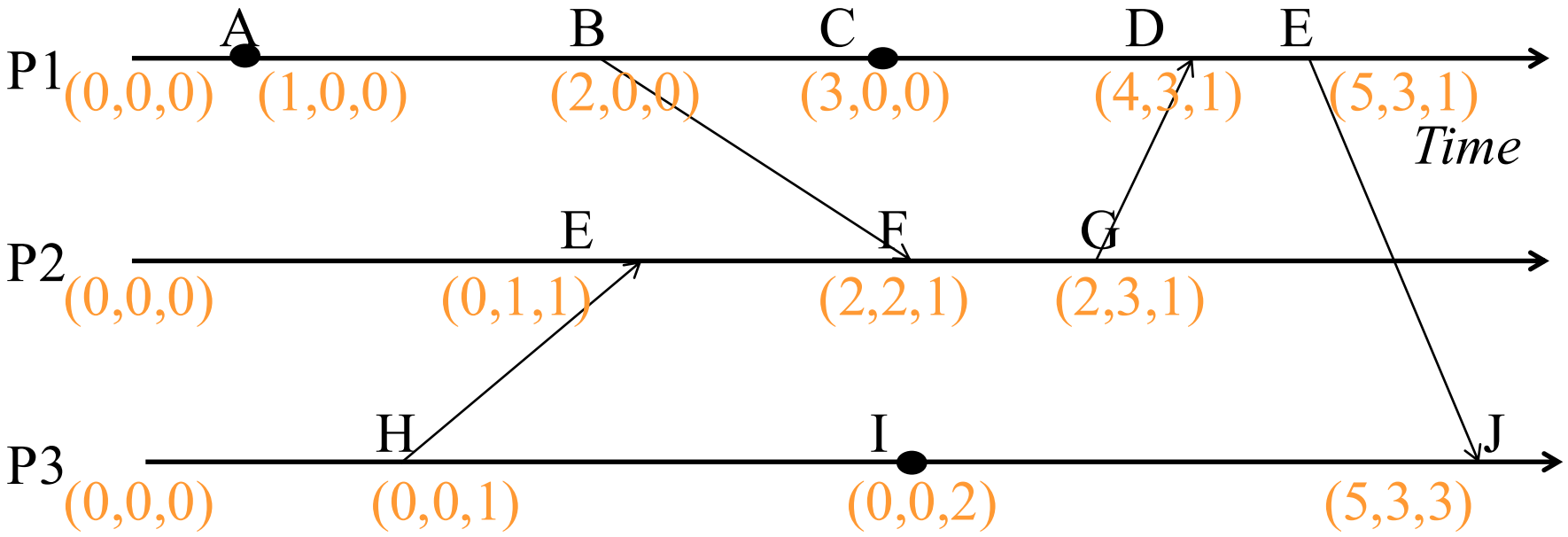

- A → B :: (1,0,0) < (2,0,0)
- B → F :: (2,0,0) < (2,2,1)
- A → F :: (1,0,0) < (2,2,1)

Obeying Causality (2)

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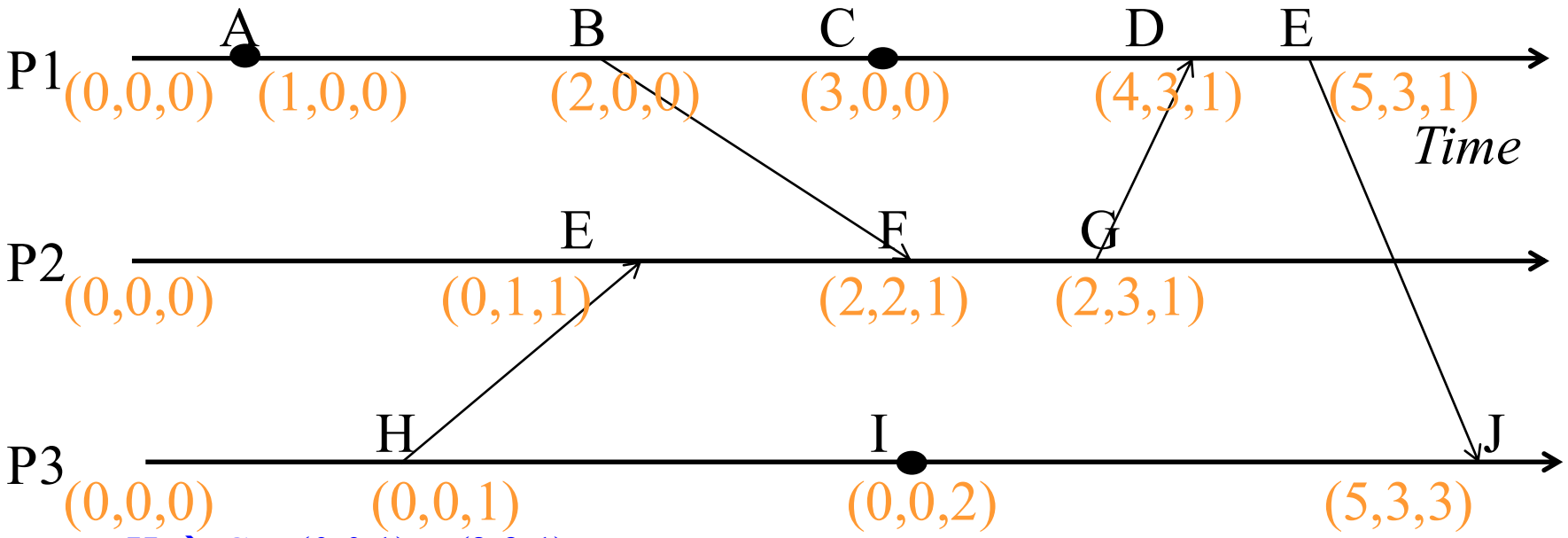

- H → G :: (0,0,1) < (2,3,1)
- F → J :: (2,2,1) < (5,3,3)
- H → J :: (0,0,1) < (5,3,3)
- C → J :: (3,0,0) < (5,3,3)

Identifying Concurrent Events

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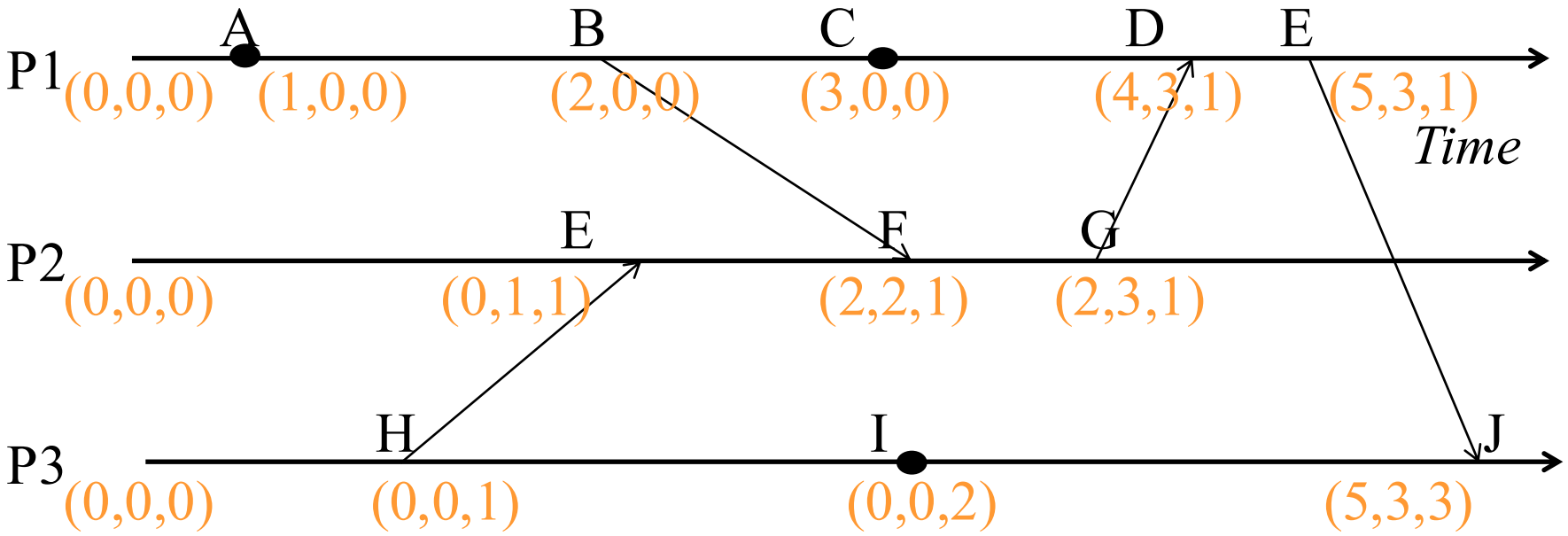

- C & F :: (3,0,0) ||| (2,2,1)
- H & C :: (0,0,1) ||| (3,0,0)
- (C, F) and (H, C) are pairs of *concurrent* events

Logical Timestamps: Summary

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- **Lamport timestamps**
  - Integer clocks assigned to events
  - Obeys causality
  - Cannot distinguish concurrent events

- **Vector timestamps**
  - Obey causality
  - By using more space, can also identify concurrent events

Time and Ordering: Summary

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- **Clocks are unsynchronized in an asynchronous distributed system**

- **But need to order events, across processes!**

- **Time synchronization**
  - Cristian's algorithm
  - NTP
  - Berkeley algorithm
  - But error a function of round-trip-time

Can avoid time sync altogether by instead assigning logical timestamps to events